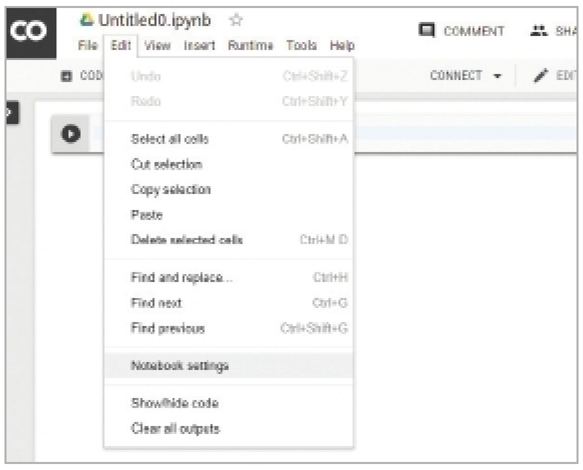

Figure 2: Accessing notebook settings

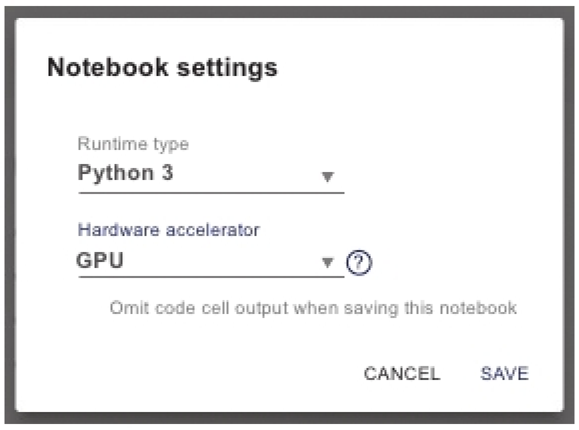

Figure 3: Hardware accelerator

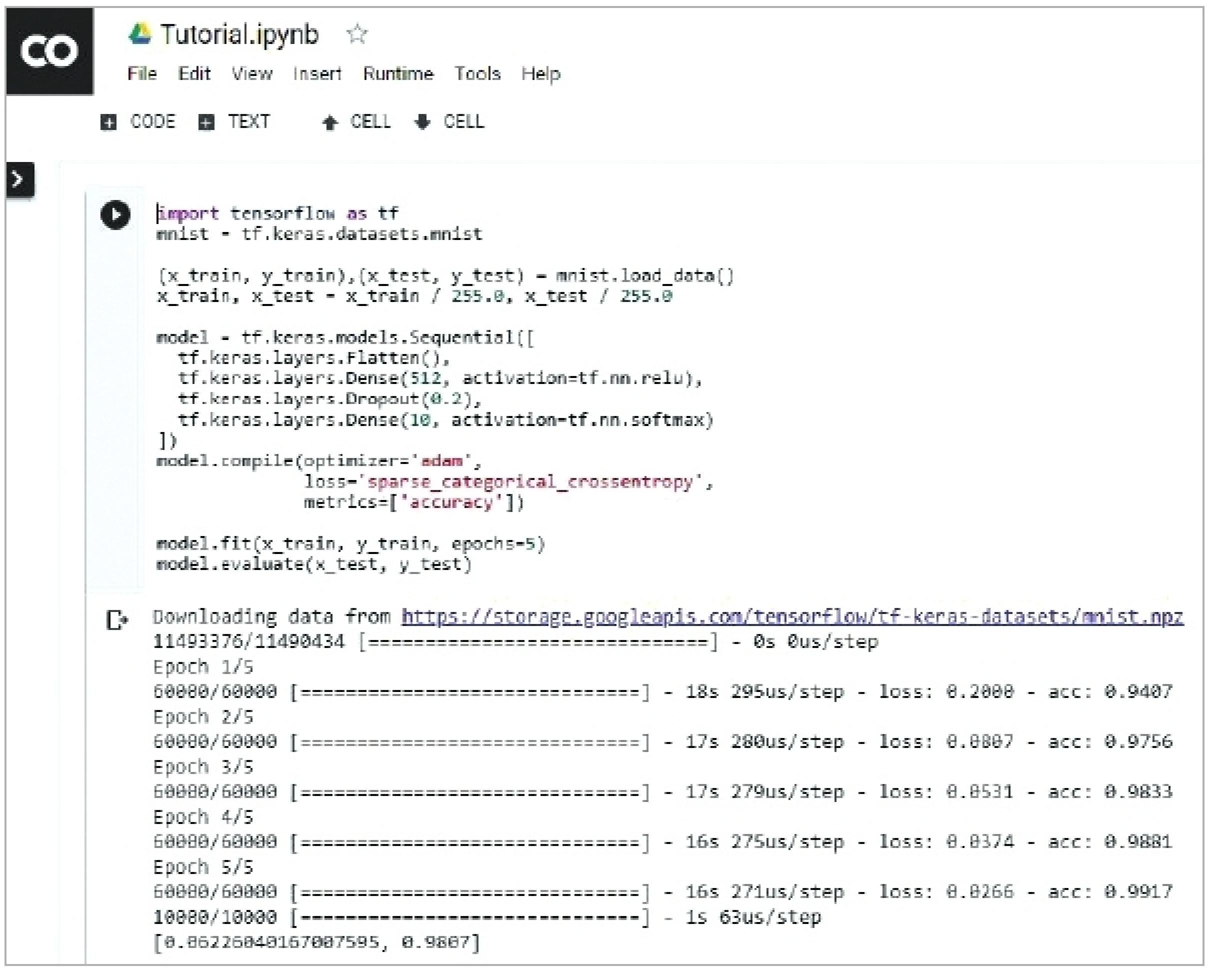

Figure 4: Mnist example

As mentioned earlier, Colaboratory is the same as any offline Python development environment that you may already be used to, thus allowing you to use packages that you normally do, such as OpenCV, TensorFlow, Keras, etc. A simple implementation of TensorFlow using Google Colaboratory is given below, starting from the official welcome page, *https://colab.research.google.com/notebooks/welcome.ipynb* .

To start off, first create your Python3 notebook in Colaboratory as shown in Figure 1.

Once you have done that, navigate to the notebook settings (*Edit > Notebook Settings*) as shown in Figure 2.

Once you have selected the notebook settings, you can now change the runtime type to Python2 or Python3 and you can also change the hardware accelerator. It is here that you will decide on whether you want use the free Nvidia Tesla K80 GPU or the TPU provided by the service (Figure 3).

Now that the environment is set up, we can finally try out some program to test it. The purpose of this article is not to take you through any specific machine learning or deep learning example. For the sake of testing Colaboratory, let's use the classic Mnist example that has been taken directly from the TensorFlow website *https://www.tensorflow.org/tutorials/*

Once you have grabbed the code, simply execute it in the notebook and you should see an output as shown in Figure 4.

Using Colaboratory, you have executed a basic classification example within minutes, without the need for any extra installations of TensorFlow (or even Python) on your computer. Thus, you can fully concentrate only on what you want to achieve and not bother about the dependencies needed for your project.

We have seen only five of the vast number libraries that are available in the open source world. These five tools will provide the reader with a strong-enough footing to push forward and explore more complex tools that are streamlined for a particular use case. I urge you to also try to learn the mathematics of any new technology that you implement as it gives you a finer understanding of the how and whys of each thing. A nice start to any theory is the Machine Learning Cheatsheet, *https://ml-cheatsheet.readthedocs.io/en/latest/index.html* which, while barely scratching the surface of the mathematics behind the implementations, gives a very clear idea about the concepts and code examples to lead you in the right direction . END

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The author takes a keen interest in graphics modelling, artificial intelligence and implementations of neural networks.